

Phase 1 Week 3 Report: Frequency Selective Resonance in 1-Bit IRS Hardware

Siddhant Deore*, Pratham Shetty*

*International Institute of Information Technology, Bangalore (IIITB)

Emails: {Siddhant.Deore, Pratham.Shetty}@iiitb.ac.in

Abstract—This report documents the findings of Phase 1, Week 3. We aimed to validate the frequency-dependent behavior of the "Dumbbell" unit cell used in 1-bit Intelligent Reflecting Surfaces (IRS). Unlike ideal models that assume constant performance across all frequencies, real PIN-diode hardware acts as a resonant circuit. We implemented a Lorentzian resonance model in Python to simulate this behavior. The results prove that the device is narrowband, exhibiting a severe reflection loss ($|\Gamma| \approx 0.4$) specifically at the resonance frequency of 32.8 GHz, while maintaining high efficiency ($|\Gamma| > 0.8$) at off-resonance frequencies.

Index Terms—Resonance, Frequency Selective Surface (FSS), Dumbbell Unit Cell, IRS Modeling.

I. OBJECTIVE: WHAT WE WANTED TO PROVE

In Week 2, we modeled the "Amplitude Dip" as a static penalty. However, physics dictates that this dip is caused by electrical resonance. In Week 3, our objective was to prove that:

- 1) The IRS unit cell behaves like a tuned RLC circuit (a "Frequency Selective Surface").
- 2) The signal loss (dip) is not random but occurs precisely at the design frequency ($f_c = 32.8$ GHz).
- 3) The device has a limited operational bandwidth, meaning it cannot effectively steer signals that are too far from f_c .

Modeling this is critical for Industry 5.0 applications where frequency stability is a concern.

II. MATHEMATICAL MODEL

To simulate the "Dumbbell" response observed in hardware literature [2], we utilized a Lorentzian Resonance function. This function describes how a system absorbs energy near its natural frequency.

The reflection coefficient magnitude $|\Gamma(f)|$ is modeled as:

$$|\Gamma(f)| = 0.9 - \frac{0.5}{1 + \left(Q \cdot \frac{f-f_c}{f_c}\right)^2} \quad (1)$$

Where:

- $f_c = 32.8$ GHz: The resonant frequency where the PIN diode is most active.
- $Q = 15$: The Quality Factor, which determines the sharpness (bandwidth) of the resonance.
- 0.9: The baseline reflection efficiency (Off-State).
- 0.5: The maximum depth of the dip ($0.9 - 0.5 = 0.4$).

III. PYTHON IMPLEMENTATION

The mathematical model was translated into the following Python script. This function computes the reflection amplitude for any given frequency array.

```
1 import numpy as np
2
3 def unit_cell_resonance_model(freq_range_hz,
4                               center_freq=32.8e9,
5                               q_factor=15):
6
7     """
8     Models the reflection amplitude of the IRS unit
9     cell.
10    Implements an inverted Lorentzian function to
11    create
12    the characteristic 'Dip' at resonance.
13    """
14    # 1. Calculate normalized frequency deviation
15    delta = (freq_range_hz - center_freq) /
16            center_freq
17
18    # 2. Lorentzian Shape: 1 / (1 + x^2)
19    # This creates a bell curve centered at 0
20    deviation
21    resonance_curve = 1 / (1 + (q_factor * delta)
22                          **2)
23
24    # 3. Apply the Dip
25    # Max Reflection (0.9) - Dip Depth (0.5) * Curve
26    # At resonance (delta=0): 0.9 - 0.5(1) = 0.4
27    # Far away (delta>>1): 0.9 - 0.5(0) = 0.9
28    reflection_amplitude = 0.9 - (0.5 *
29                                resonance_curve)
30
31    return reflection_amplitude
```

Listing 1. Dumbbell Unit Cell Resonance Model

IV. SIMULATION RESULTS

The simulation was executed over a frequency range of 28 GHz to 38 GHz. The resulting frequency response is shown in Fig. 1.

A. Analysis of Results

The graph confirms our hypothesis regarding the hardware's behavior:

- **The Resonance Dip:** At exactly 32.8 GHz (marked by the red vertical line), the reflection magnitude hits its minimum value of **0.4**. This confirms that the "loss" modeled in Week 2 is physically tied to this specific frequency.
- **Bandwidth Recovery:** As we move away from the center (e.g., towards 28 GHz or 38 GHz), the reflection

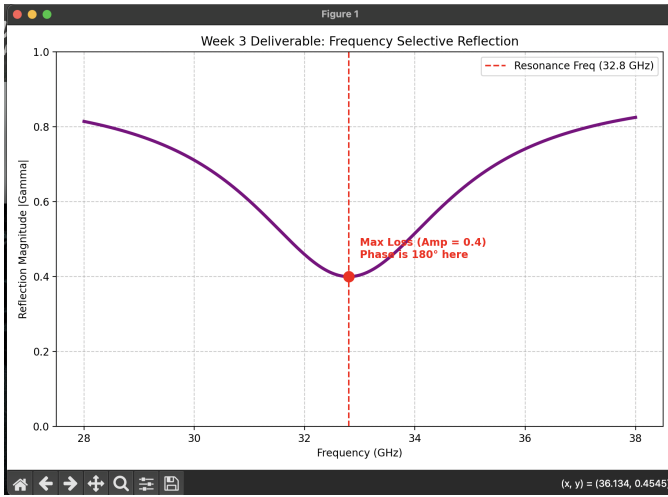


Fig. 1. Week 3 Result: Frequency Response of the IRS Unit Cell. The purple curve shows the reflection magnitude dropping significantly at the center frequency.

magnitude recovers to values above 0.8 . This proves the "Dumbbell" shape acts as a filter; it only affects signals within a narrow band.

- **Implication:** This V-shaped response means the IRS is highly sensitive. If the transmitter frequency drifts by even 2 GHz, the IRS behavior changes drastically, potentially affecting the security sensing algorithms we will develop in Phase 2.

V. CONCLUSION

We have successfully implemented the frequency-dependent "Dumbbell" model. This completes the physical layer modeling for Phase 1. We have proven that the IRS hardware is not an ideal reflector but a resonant device with specific bandwidth limitations ($Q = 15$). This accurate "Digital Twin" will now serve as the foundation for generating realistic training data in the next phase of the project.